

Young Children and ChatGPT: Parents' Use of ChatGPT in Parenting

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ABSTRACT

The increased use of LLMs such as ChatGPT introduced new learning opportunities for educators and different age groups of learners in formal education. However, it is unknown whether parents and young children are also adopting ChatGPT in informal learning environments. Using the Technology Acceptance Model (TAM) as a framework, this study surveyed 74 parents in the U.S. with young children between five and eight years old and found both parent and child use of ChatGPT. Hierarchical linear regression models were employed and revealed that parents' positive attitudes toward ChatGPT and parental encouragement are significantly related to parents' use of ChatGPT, and children's use of ChatGPT is significantly influenced by parental encouragement. As one of the first known studies to examine ChatGPT use in parenting, this study contributes to understanding the role of LLMs as AI tools for parenting in home learning environments.

CCS CONCEPTS

• Human-centered computing; • Collaborative and social computing; • Empirical studies in collaborative and social computing;

KEYWORDS

ChatGPT, Young Children, Parenting, Educational Technology, Home Learning Environment

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1 INTRODUCTION

The increasing integration of Artificial Intelligence (AI) tools like Large Language Models (LLMs) and Generative Pre-trained Transformer (GPT) technology has led to drastic changes in education from early childhood to adulthood [12, 14, 17, 19, 23, 25, 30, 31, 38].

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Numerous studies have explored the benefits and opportunities for integrating LLMs such as ChatGPT to help learners, students, and teachers in formal educational settings from K-12 schools [12, 24, 25] to university classrooms [27, 31, 38]. Aside from known challenges with AI tools such as trustworthiness and accuracy [42], previous research has reported that AI tools are beneficial to increasing learners' conceptual knowledge about mathematics, science, engineering, and language and literacy learning [2, 5, 15, 28, 29, 41]. In a recent review, Su & Yang [41] stated that multiple stakeholders benefit from integrating AI into early childhood education (ECE), including learners, teachers, parents, and caregivers. ChatGPT, however, as an AI tool, has been examined as a valuable educational tool among educators and students in school settings, but not among other key stakeholders such as parents and young children in the home learning environment [18].

Home learning environments and early learning experiences have long been considered to play a critical role in children's development, school readiness, and academic performance [35, 40, 49]. In the home learning environments, young children develop their conceptual knowledge and literacy skills with critical influence from parents, as parents are key informants who support children in developing their social and learning skills [20, 22, 32, 35, 36]. Therefore, it is essential to explore how LLMs such as ChatGPT support parents in assisting with their young children's learning at home. This survey study aimed to explore whether the parent and child use of ChatGPT is influenced by parents' awareness of ChatGPT's usefulness, their positive attitudes, and parental encouragement regarding their children's use. The study seeks to answer the following two research questions:

- (1) What are the relationships among parents' awareness of ChatGPT's usefulness, positive attitudes, parental encouragement, as well as parent and children's use of ChatGPT?
- (2) Do parents' awareness of ChatGPT's usefulness, positive attitudes, and parental encouragement significantly predict their personal, parenting, and children's use, respectively?

2 RELATED WORK

2.1 Parenting Behaviors and Activities When Using LLMs

In the past few years, various researchers have explored the use of AI tools for parenting, such as using AI-enabled systems to assist parent-child storytelling experience [44] and to promote effective parenting skills such as praising [8–11]. Despite the extensive adoption of LLMs for language and academic learning, only a few explored the usefulness of LLMs in parenting. Luo et al. [21]

interviewed early childhood education experts from China and the United States (U.S.) and found that one role of ChatGPT in parenting is to serve as an on-call facilitator for educators and caregivers, which leads to new opportunities and challenges related to AI literacy and AI social interaction [20, 46]. Similar findings were also reported by McFayden et al. [23] through the evaluation of ChatGPT-4 for parents with children with autism when they sought information about autism. Authors observed that ChatGPT can provide parents with accurate answers but only a little actionable advice and limited or inaccurate references and hyperlinks. Given the fact that some parents have begun to explore AI tools for parenting tasks, it is important to further characterize parents' interactions with LLMs such as ChatGPT to support their children's learning [20].

2.2 LLMs as AI Tools for Young Children

Many empirical studies have demonstrated the usefulness of LLMs for supporting academic and language learning [1, 3, 16, 19, 43, 45]. Most prior research on ChatGPT has focused on classroom learning in English or other European languages [18]. Very few studies explored the use of ChatGPT in informal learning settings such as home, which is a critical learning environment that is beneficial to young children. Jauhiainen & Guerra [12] investigated the utility of generative AI-modified learning materials (e.g., text, figures, exercises) in a class lesson with 110 primary school students from eight to fourteen years old (grade four to six) who received Spanish as the primary instructional language. Researchers reported that ChatGPT 3.5 enables the personalization of learning materials to different levels of knowledge and motivates student learning and skill development in four key subjects, including social science, math, English, and Spanish. Murgia et al. [7] explored fourth-grade students between ages nine and ten using ChatGPT to learn

the primary school curriculum. Results suggested opportunities for using ChatGPT to support children with various language and cognitive skills and adapting to individual learning needs to improve teachers' instructional needs. Given that the nature of LLMs can be used to improve information seeking and discovery, it will likely lead to different language and early literacy learning levels in young children before primary school. To understand learning activities prior to school age, it is critical to examine how caregivers such as parents provide informal education during their parenting activities using LLMs at home. It is unknown how parents adopt ChatGPT in parenting and what factors impact their use, especially among parents of young children between the ages of five and eight [13, 23]. More empirical research can help researchers understand what factors influence parents' use of ChatGPT as a novel AI parenting tool and how the new generation of youth interact with LLMs at home [40].

3 METHODS

3.1 Study Design

The study employed a non-experimental, cross-sectional quantitative research design to investigate the use of ChatGPT by parents and children at a single point in time [37]. This research is part of a broader study into the home learning environment of young children. The Technology Acceptance Model (TAM), one of the most widely used frameworks to examine the acceptance of emerging educational technology [4, 6, 17, 33, 34, 53, 54], was adopted to explore the experiences of parents and children using ChatGPT in the home environment. According to TAM, Perceived Ease of Use (PEU), Perceived Usefulness (PU), Attitude Towards Using (ATU), and Behavioral Intention of Use (BI) influence the Actual Use (AU) of emerging technologies [5, 7, 26, 47]. Focusing on four of five TAM constructs: PU, ATU, BI, and AU, this study hypothesized that

Table 1: Survey items and TAM constructs.

Survey Item	TAM Construct	Definition	Label
I am aware of ChatGPT and its capabilities	PU	A construct represents parent's awareness of ChatGPT's usefulness	Awareness of Usefulness
I believe that ChatGPT has the potential to positively transform education	ATU	A construct refers to parent's attitude toward ChatGPT	Positive Attitude
I actively encourage and guide my child in using ChatGPT	BI	A construct refers to a parent's intention toward their child's use of ChatGPT	Parental Encouragement
I have concerns about the potential impact of ChatGPT and therefore set limitations on my child's usage			
I have personally interacted with ChatGPT	AU	A factor represents parent's personal use of ChatGPT	Personal Use
ChatGPT aids me in various parenting tasks and challenges	AU	A factor related to parenting purposes use of ChatGPT	Parenting Use
I utilize ChatGPT to assist my child's learning process			
I leverage ChatGPT to enhance my child's language learning abilities			
My child has also used ChatGPT	AU	A factor representing child use of ChatGPT	Child Use

parents' awareness of ChatGPT's usefulness (PU), their positive attitude toward ChatGPT (ATU), and their encouragement of their children's use of ChatGPT (BI) would impact their actual use (AU) of ChatGPT. AU was further categorized into three dimensions: parent's personal use, parenting use, and child use. Since children's early learning experiences are formed by modeling their parent's behavior in the home learning environment [40, 49, 64], the inclusion of the parent's personal use would allow the study to investigate whether a parent's personal use impacts their child's use of ChatGPT. Parenting use was further expanded into three areas: use for parenting purposes, use for children's general learning, and use for children's language learning, aiming to capture the usage patterns in the parenting context. PEU was excluded from this study because parent-involved educational technology research typically focuses on parents' attitudes toward their children's use [55, 57–61], and PEU is commonly employed to examine parental acceptance of specific educational technologies such as school-introduced software [56, 65, 66]. Moreover, at the time of survey design, ChatGPT had just been released in various versions (e.g., web vs. app, freemium vs. pro), bringing challenges in capturing its usability. Grounded in TAM, a nine-item survey was developed, utilizing a five-point Likert scale ranging from 1 for "never/strongly disagree" to 5 for "very often/strongly agree." The survey items, along with TAM constructs and their corresponding definitions as applied in this study, are described in Table 1.

3.2 Participants

Using the snowball sampling method, an IRB-approved online survey was administered via Qualtrics from August 2023 to January 2024, with initial participation from 83 families with children aged five to eight enrolled in U.S. schools and fluent in English nationwide. As this research is part of a more extensive study that involves meetings with children following their survey completion, the children's ages were therefore validated without fraudulent reports.

3.3 Data Analysis

Data was cleaned and analyzed using SPSS 29.0. After dropping incomplete responses, 74 valid datasets were retained for analysis. Two composite variables: parental encouragement and parenting use, were created by computing the mean value of observing variables for data analysis. The hierarchical linear regression model was employed as it is commonly used in studies exploring parental involvement and parent-child interaction in parenting contexts [39, 48], such as the Home Literacy Model [35]. The current study applied hierarchical linear regression to examine the predictive power of parents' awareness of usefulness, positive attitude, and parental encouragement to parent's and child's use of ChatGPT.

4 RESULTS

4.1 Demographic Characteristics

The 74 survey responses collected nationwide included 59.5% of parents from New Jersey and 13.5% of parents from New York. Most caregivers who responded to the survey are females (84.7%, $N=60$) and mothers (83.1%, $N=59$). Most parents identify as non-Hispanic (97.1%, $N=68$) and Asian (83.1%, $N=59$), and about half are in the 36–40 age group (46.5%, $N=33$). A significant portion

of the parents reported a high income level, with 63.2% ($N=43$) earning more than \$200,000 per year. The children in the study have an average age of 77.8 months ($SD=9.334$). Most are non-Hispanic (95.8%, $N=68$) and Asian (76.1%, $N=54$). 69% of children are in elementary school ($N=49$), and 74.3% ($N=52$) have siblings. Bi/multilingualism is common, with 83.1% ($N=59$) of the children from bilingual or multilingual families.

4.2 Descriptive Statistics

Results from descriptive statistics suggested that 71.6% of parents reported personal use of ChatGPT, indicating a moderate to high level of usage (Mean=3.53, $SD=1.53$); 52.7% of parents reported using ChatGPT for various parenting purposes, including seeking parenting strategies and assisting their children in general and language learning (Mean=2.23, $SD=.85$). Additionally, 8.1% of children have used ChatGPT as reported by their parents (Mean=1.28, $SD=.69$). These results suggest an active use of ChatGPT among parents and children in the home learning environment and parenting context.

Table 2: Descriptive statistics of ChatGPT use

	Percentage	Mean	SD
Parent's Personal Use	71.6	3.53	1.53
Parent's Parenting Use	52.7	2.23	.85
Child Use	8.1	1.28	.69

4.3 Correlation Analysis

Pearson correlation analysis was employed to explore the correlation between each variable. According to the results (Figure 1), parent's personal use of ChatGPT is positively and significantly correlated with their awareness of ChatGPT's usefulness ($r=.640$,

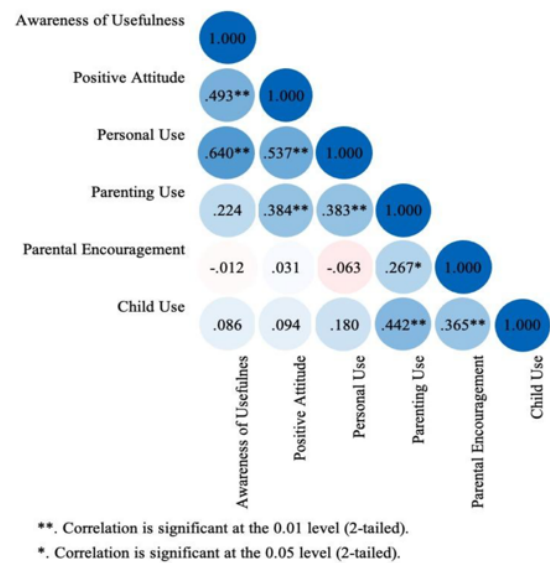


Figure 1: Correlation Matrix of ChatGPT Use

Table 3: Hierarchical Regression Analysis for Parent's Personal Use

Model 1: Personal Use	R^2	R^2_{adj}	ΔR^2	F	F-Change
Awareness of Usefulness	.409	.401	.409	49.925***	49.925***
Positive Attitude	.474	.460	.065	32.033***	8.760**
Parental Encouragement	.479	.456	.004	21.430***	.592

Table 4: Hierarchical Regression Analysis for Parenting Use

Model 2: Parenting Use	R^2	R^2_{adj}	ΔR^2	F	F-Change
Awareness of Usefulness	.050	.037	.050	3.788	3.788
Positive Attitude	.149	.125	.099	6.204**	8.239**
Parental Encouragement	.214	.181	.066	6.371***	5.857*

Table 5: Hierarchical Regression Analysis for Child Use

Model 3: Child Use	R^2	R^2_{adj}	ΔR^2	F	F-Change
Awareness of Usefulness	.007	-.006	.007	.533	.533
Positive Attitude	.011	-.017	.004	.390	.254
Parental Encouragement	.143	.107	.132	3.905*	10.825**

$p < .001$), their positive attitudes towards ChatGPT ($r = .537$, $p < .001$), and their parenting use ($r = .383$, $p < .001$). Furthermore, parenting use is positively and significantly correlated with parent's positive attitude ($r = .384$, $p < .001$), personal use ($r = .383$, $p < .001$), and parental encouragement ($r = .267$, $p = .022$), as well as to child use ($r = .442$, $p < .001$). Additionally, child use of ChatGPT demonstrates a significant positive relationship with parental encouragement ($r = .365$, $p = .001$), alongside its significant correlation with parenting use.

4.4 Hierarchical Regression Models for Personal, Parenting, and Child Use

Three hierarchical regression models were employed to examine how much of the variances of the use can be explained by three predictors: parent's awareness of usefulness, their positive attitude, and parental encouragement. Following the order in which the factors in the TAM model affect the usage [17, 52], awareness of usefulness was put in the first level, followed by the positive attitude as the second, and parental encouragement in the third.

4.4.1 Personal Use. Model 1 (Table 3) suggested that parents' awareness of the usefulness significantly predicted their personal use, explaining about 40.9% of the variance. While positive attitude increased the model's explanatory power, contributing a significant additional 4.7% to the variance. Parental encouragement showed a limited and non-significant predictive power on personal use.

4.4.2 Parenting Use. In Model 2 (Table 4), awareness of usefulness demonstrated a relatively modest explanatory power ($R^2 = .050$, $F = 3.788$), suggesting a limited impact on parenting use of ChatGPT. However, the addition of positive attitude significantly enhanced the model, accounting for an additional 9.9% of the variance

($\Delta R^2 = .099$, $F\text{-Change} = 8.239$, $p < .01$). Parental Encouragement further contributed to the model, increasing the R^2 to .214 and adding a significant 6.6% to the variance explained. In total, three predictors explained 21.4% of the variance in parenting use.

4.4.3 Child Use. In Model 3 (Table 5), three predictors explained a total of 14.3% of the variance in child use. Awareness of usefulness ($R^2 = .007$, $F = .533$) and positive attitude ($\Delta R^2 = .004$, $F\text{-Change} = .254$) showed a limited and non-significant influence on the variance of child use. In contrast, parental encouragement served as the strongest significant predictor in the model and explained an additional 13.2% of the variance of child use ($\Delta R^2 = .132$, $F\text{-Change} = 10.825$, $p < .01$). This highlights that parental encouragement plays a significant role in influencing children's use of ChatGPT.

5 DISCUSSION

Utilizing descriptive statistics, correlation, and regression analyses, survey results from 74 families suggested that 71.6% of parents have personally interacted with ChatGPT, and over half of the parents reported using ChatGPT in parenting activities. Approximately 8.1% of parents reported that their children have used ChatGPT. This finding provides one of the first pieces of evidence about five- to eight-year-old young children's use of LLMs such as ChatGPT in informal learning settings, filling in the knowledge gaps in the current research field. Furthermore, study results exhibit notable patterns among personal use, parenting use, and child use through correlation analysis. The significant positive correlation between parent's personal use and parenting use suggests that a certain portion of parent's ChatGPT use is related to parenting activities, which is consistent with prior work on how parents leverage AI

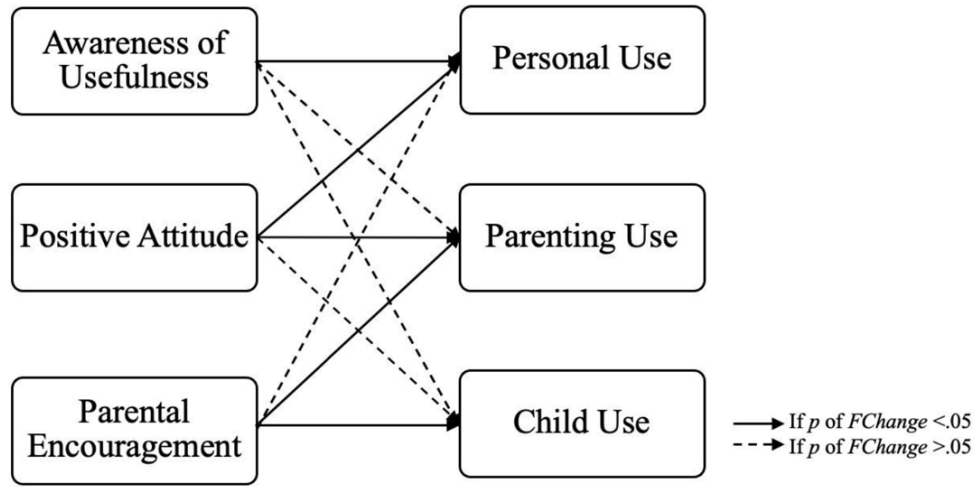


Figure 2: Conceptual Model of Hierarchical Regression Findings

tools and LLMs in parenting activities [8–10, 21, 24, 44]. Additionally, child use of ChatGPT demonstrates a significant positive correlation with parents' parenting use but not with their personal use. While previous studies have documented parents using AI tools for parenting in an interactive way with their children [44], in the present study, it remains unclear whether children's use of ChatGPT is an independent activity or collaborated with their parents.

Three hierarchical regression models were presented in Figure 2 as a conceptual model of findings, with arrows indicating significant relations. Based on survey results, awareness of usefulness and positive attitude appear to significantly influence parent's personal use of ChatGPT, and parenting use of ChatGPT appears to be affected by positive attitude and parental encouragement. Such findings indicated that when adopting emergent technology in parenting, fostering a positive attitude in parents might be crucial. Nevertheless, while parenting use is significantly related to their positive attitude and encouragement toward their children, child use appears to be solely influenced by parental encouragement. This finding is consistent with related works on the positive impact of parental encouragement in engaging children with various educational technologies [57, 59, 66], indicating that such encouragement plays an equally critical role in involving young children with AI tools. Although parental encouragement significantly predicts child use, it is noteworthy that the predictive power is mild, with 13.2% of the variance in child being explained by parental encouragement. This finding implies that other factors mentioned in the related work such as age [51], technology accessibility [50], or motivation [17] might also impact child use. Exploring demographic factors, expanding the frameworks, involving open-ended questions, and conducting follow-up interviews would lead to a further understanding about children's use of AI tools in the home learning environment.

6 CONCLUSION, LIMITATION, AND FUTURE WORK

This study is one of the first known studies to survey using experiences of ChatGPT for parents and young children. Using the TAM framework as a theoretical grounding, survey results from 74 families in the U.S. with young children who are between five and eight years old reported an active use of ChatGPT in the home setting, revealing that awareness of usefulness, positive attitude, and parental encouragement influence parent's and child's use of ChatGPT in different significance levels. It suggested positive attitude and parental encouragement significantly impact parents' use of ChatGPT, and child use of ChatGPT is significantly affected by parental encouragement. This study has several limitations that could be addressed in future research. First, the recruitment of participants using snowball sampling through personal networks raises concerns about homogeneity. The majority of participants were from families of high socioeconomic status, and there was a higher proportion of bilingual families, which may not be representative of the broader U.S. demographics. Incorporating more participants from diverse backgrounds would be advantageous in increasing the sample size and enhancing the study's representativeness. Secondly, given that this is a self-reported survey by parents from a broader comprehensive study, it does not capture the parents' perceived ease of use (PEU) or detail how children interacted with ChatGPT, whether independently or in collaboration with their parents. Future research could adopt the complete TAM framework, incorporating open-ended survey questions and interviews to assess the usability and usage scenarios of ChatGPT as mentioned by prior work [62, 63]. This approach would provide deeper insights into children's interactions with ChatGPT and its specific contributions to their learning outcomes. In summary, this study provides timely insights into the use of generative AI tools such as LLMs in supporting parenting and enhancing learning experiences for young children within home environments.

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